

WOOKYOUNG KIM, Ph.D.

Senior Researcher | Thermal-Fluid Engineering

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PROFESSIONAL SUMMARY

Thermal-fluid engineer with 5+ years of R&D experience at Korea's national research institute (KIMM). Specialized in thermal management solutions for AI data centers, hydrogen energy systems, and heat pump technology. Extensive hands-on experience in thermal system design, experimental facility construction, performance testing, and data analysis. Spearheading development of multiple engineering software tools (Python/FastAPI/React Native). Track record: 17+ journal papers, 17 patents, 8 registered software programs, 5 technology transfers to industry.

CORE COMPETENCIES

Thermal Management	Energy Systems	System Design	Testing & Analysis
Data center cooling	Heat pump (vapor./ads.)	Heat exchanger design	Performance testing
Immersion / DLC / Jet cooling	Hydrogen vaporizer (PCHE)	Thermal loop construction	2-phase flow visualization
Electronics thermal design	Low-GWP refrigerant R&D	High-pressure cryogenic sys.	VLE measurement & EOS

EXPERIENCE

Senior Researcher | Korea Institute of Machinery and Materials (KIMM) | Mar 2021 - Present

Korea's national research institute for machinery and materials engineering

- PI: Jet-enhanced immersion cooling for next-gen data center cooling
- PI: KIMMPROP - Thermophysical property iOS/Android app development
- Lead: PCHE design & testing for liquid hydrogen vaporizers (down to -220°C, 100 MPa)
- Lead: Immersion cooling waste heat utilization and thermal management
- Delivered 5 technology transfers to industry partners
- Developed 8 registered engineering software programs (Korea Copyright Commission)

EDUCATION

B.S.	Mechanical Engineering	KAIST	2011 - 2015
M.S.	Mechanical Engineering	KAIST	2015 - 2017
Advisor: Prof. Sung Jin Kim			
Ph.D.	Mechanical Engineering	KAIST	2017 - 2021
Dissertation: Study on a Thermal Network-based Model for Predicting the Thermal Resistance of Pulsating Heat Pipes			
Advisor: Prof. Sung Jin Kim			

KEY ACHIEVEMENTS

17+	17	8	5
Journal Papers (10 SCI + 7 KCI)	Domestic Patents	Registered Software Programs	Technology Transfers to Industry

SELECTED PUBLICATIONS

- W. Kim et al., "Freezing Phenomenon in PCHE for Cryogenic LH2 Vaporizer," Appl. Therm. Eng. 273 (2025)
- W. Kim and S.J. Kim, "Fundamental issues about pulsating heat pipes," J. Heat Transfer - ASME 143 (2021)
- W. Kim and S.J. Kim, "Flow behavior effect on pulsating heat pipes," Int. J. Heat Mass Transfer 149 (2020)
- W. Kim and S.J. Kim, "Reentrant cavities on pulsating heat pipe," Appl. Therm. Eng. 133 (2018)
- J.S. Kim, W. Kim et al., "Pool boiling of ammonia outside enhanced tubes," Appl. Therm. Eng. 247 (2024)

TECHNICAL SKILLS

Software: Python, JavaScript/TypeScript, C/C++, FastAPI, React, React Native, Git, Docker, Linux

CAD / CAE: INVENTOR, SOLIDWORKS, COMSOL Multiphysics, ANSYS FLUENT, CoolProp, REFPROP

AI / ML: PyTorch, JAX, PINN

Experiment: Thermal loop design & construction (1-phase / 2-phase), Low GWP Refrigerant system, 2-phase flow & heat transfer measurement, High-pressure testing (100 MPa), Flow visualization

PROFESSIONAL ACTIVITIES

- Council Member**, Korean Society of Mechanical Engineers (KSME) [2026.04 - Present]
- Committee Member - Thermal Management / Cooling**, SAREK Data Center Technology Committee [2026.03 - Present]
- Regular Member**, Society of Air-Conditioning and Refrigerating Engineers of Korea (SAREK) [2022.04 - Present]
- Regular Member**, Korean Society of Mechanical Engineers (KSME) [2021.05 - Present]